State simulator dfference between Cirq and TFQ

https://github.com/tensorflow/quantum/issues/235

ROW 29

State simulator dfference between Cirq and TFQ #235

New issue

Closed #236

therooler opened on May 7, 2020

Environment:
Python 3.7
tensorflow = 2.1.0
tensorflow-quantum = 0.3.0
Cirq = 0.7.0

Issue: When running larger (N = 12) variational circuits `cirq.Simulator()` and `tfq.layer.State` output states that do not agree. When using this in VQE to find the ground state of some Hamiltonian, after convergence, the Cirq state has overlap ~1 with the ground state whereas the TFQ state does not. This problem gets worse as N increases.

Reproducible example
We run a depth 4 circuit with ZZ gates between nearest neighbors on an open chain of length N.
With N=8, the overlap between the two states is 1. But when we go to a chain of length 12, this is no longer the case.

```
import cirq
import tensorflow_quantum as tfq
import tensorflow as tf
import numpy as np
import sympy


def unique_name():
    """
    Generator to generator an infinite number of unique names

    Yields: Strings of the form 'theta_<number>'

    """
    num = 1
    while True:
        yield 'theta_' + str(num)
        num += 1


def issue(nqubits):
    depth = 4

    name_generator = unique_name()

    qubits = cirq.GridQubit.rect(nqubits, 1)
    circuit = cirq.Circuit()

    control_params = []

    circuit.append(cirq.H.on_each(qubits))
    for _ in range(depth):
        name = next(name_generator)
        control_params.append(name)
        symbol = sympy.Symbol(name)

        circuit.append(cirq.ZZ(qubits[i], qubits[i + 1]) ** symbol for i in
                       range(0, nqubits - 1, 2))
        circuit.append(cirq.ZZ(qubits[i], qubits[i + 1]) ** symbol for i in
                       range(1, nqubits - 1, 2))
        circuit.append(cirq.ZZ(qubits[nqubits - 1], qubits[0]) ** symbol)

    initial_value = np.random.rand(1, len(control_params))

    params_dict = dict(zip(control_params, initial_value[0]))
    resolver = cirq.ParamResolver(params_dict)
    phi_cirq = cirq.Simulator().simulate(circuit, resolver).final_state

    state_layer = tfq.layers.State()
    model_params = tf.Variable(initial_value=initial_value)
    # phi_tfq = \
    #     state_layer(circuit, symbol_names=control_params,
    #                 symbol_values=initial_value)[
```

Assignees
No one assigned

Labels
No labels

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
Precision fix
tensorflow/quantum

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Participants

```
    params_dict = dict(zip(control_params, initial_value[0]))
    resolver = cirq.ParamResolver(params_dict)
    phi_cirq = cirq.Simulator().simulate(circuit, resolver).final_state

    state_layer = tfq.layers.State()
    model_params = tf.Variable(initial_value=initial_value)
    # phi_tfq = \
    #     state_layer(circuit, symbol_names=control_params,
    #                 symbol_values=initial_value)[
    #         0].numpy()
    phi_tfq = \
        state_layer(circuit, symbol_names=control_params,
                    symbol_values=model_params)[
            0].numpy()

    overlap = np.abs(np.conj(phi_cirq) @ phi_tfq)
    return overlap


if __name__ == "__main__":
    print(issue(8))
    print(issue(12))

>>> 0.9999997
>>> 0.5101427
```

Is our implementation incorrect? Or have we encountered a bug?

MichaelBroughton on May 7, 2020 Collaborator ···

Was able to recreate on my end, tried using `backend=cirq.Simulator()` as well, which did bring things back to 0.99. My plan now is to review all of the changes that might have been made between our C++ version of qsim and the reference version https://github.com/quantumlib/qsim . Will report back with findings soon.

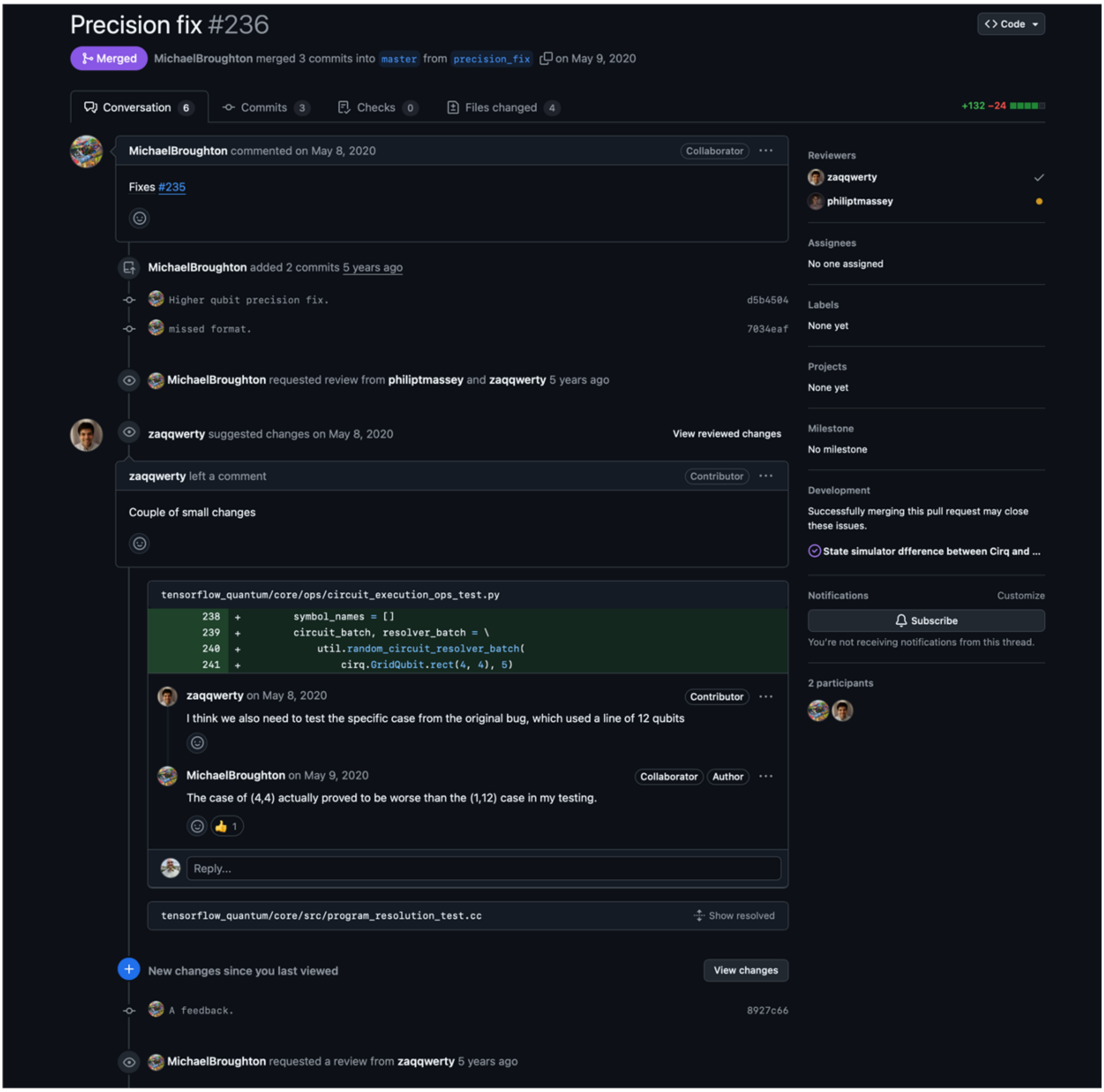
MichaelBroughton mentioned this on May 8, 2020

Precision fix #236

MichaelBroughton closed this as completed in #236 on May 9, 2020

jaeyoo added a commit that references this issue on Mar 30, 2023

`Merge pull request tensorflow#235 from quantumlib/thread-recommend` … Verified 4705b09


Precision fix #236
Merged
MichaelBroughton merged 3 commits into master from precision_fix on May 9, 2020
Code
Conversation 6
Commits 3
Checks 0
Files changed 4
+132 −24
MichaelBroughton commented on May 8, 2020
Collaborator
Fixes #235
MichaelBroughton added 2 commits 5 years ago
Higher qubit precision fix. d5b4504
missed format. 7034eaf
MichaelBroughton requested review from philiptmassey and zaqqwerty 5 years ago
zaqqwerty suggested changes on May 8, 2020
View reviewed changes
zaqqwerty left a comment
Contributor
Couple of small changes
tensorflow_quantum/core/ops/circuit_execution_ops_test.py
238 + symbol_names = []
239 + circuit_batch, resolver_batch = \
240 + util.random_circuit_resolver_batch(
241 + cirq.GridQubit.rect(4, 4), 5)
zaqqwerty on May 8, 2020
Contributor
I think we also need to test the specific case from the original bug, which used a line of 12 qubits
MichaelBroughton on May 9, 2020
Collaborator
Author
The case of (4,4) actually proved to be worse than the (1,12) case in my testing.
1
Reply...
tensorflow_quantum/core/src/program_resolution_test.cc
Show resolved
New changes since you last viewed
View changes
A feedback. 8927c66
MichaelBroughton requested a review from zaqqwerty 5 years ago
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zaqqwerty
philiptmassey
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Labels
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Projects
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